

EC421: Problem Set 7

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Problem 1

Consider an exporter from Germany to the U.S. with the constant marginal costs equal to €1. The initial exchange rate between the countries is 1\$/€. Calculate the pass-through of the exchange rate into the price of the German good paid by the consumers in the U.S. in each of the following cases:

- A) The exporter charges a constant markup and there are no distribution costs.
- B) The distribution costs in the U.S. are \$0.5 and are independent from the exchange rate.
- C) There are complementarities in price setting equal $\alpha = 0.4$ and the prices of competitors in the U.S. are constant in dollars $P = 1$.
- D) Other firms in the U.S. also have $\alpha = 0.4$ and their marginal costs are completely stable in dollars. The share of the German producer is 30% of the U.S. market.

Problem 2

Choose a country and download (quarterly) data on its trade-weighted exchange rate, exchange rate against the dollar, exchange rate against the euro, import price index, export price index, real imports and real exports. Take logs and first differences. Regress prices and quantities on the exchange rates (separately and simultaneously). Discuss the results. Prepare slides showing the raw data and the estimated pass-through coefficients.